Effects of U.S. Export Controls on Chinese Firms

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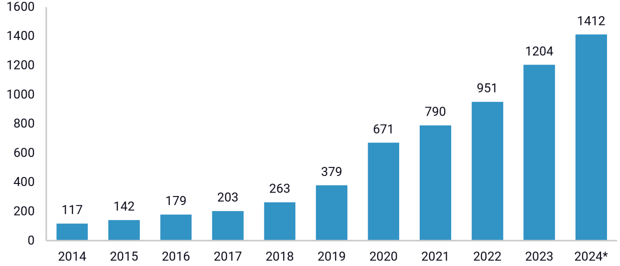
Motivation: Shifting US Policy

Heightened competition with China is driving a shift in U.S. economic policy priorities.

FIGURE 1
Number of Chinese entities designated under US sanctions and red-flag lists



Source: Rhodium Group. *Number of Chinese entities as of July 1, 2024.

Estimating US policy impacts on China is uniquely challenging due to: **Data Limitations** and its **Large Non-Market Economy**

Recent Economic Policy Changes Targeting China

- BIS Entity List
- Section 232/301 & AD/CVD Tariffs
- USG Procurement Bans: Sec. 889, FCC "Rip and Replace", NS-CMIC List, 1260H CMC List
- Uyghur Forced Labor Prevention Act
- Expanded CFIUS scrutiny and SDN listings

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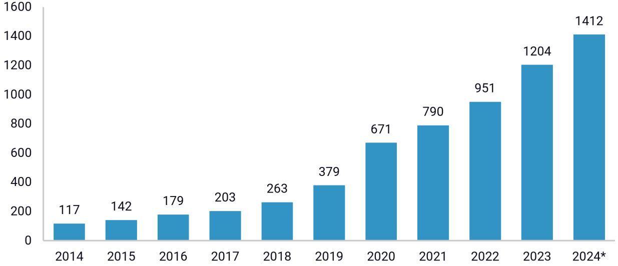

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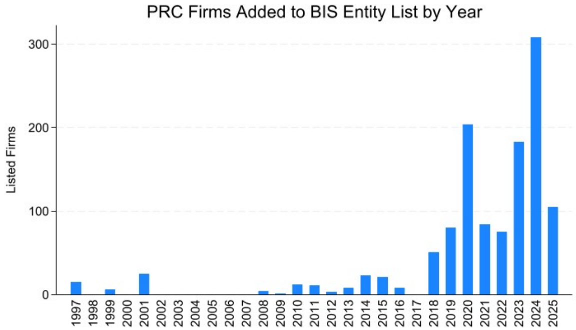
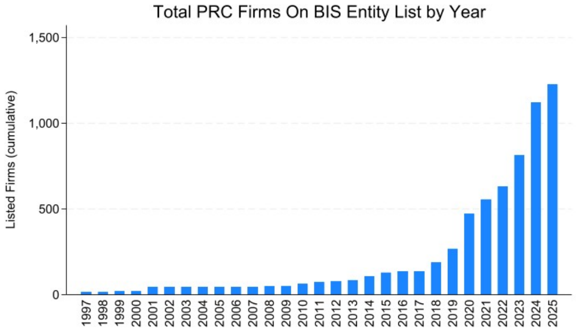
BIS Entity Listings of PRC Firms Over Time





Note: Data from International Trade Administration's Consolidated Screening List (CSL).

What is the BIS Entity List?

- The **Bureau of Industry and Security (BIS) Entity List** is a firm based U.S. export control tool that restricts the transfer of sensitive goods, software, and technology to designated foreign entities.

- Listed firms are subject to **heightened licensing requirements**. In practice, firms face a "**presumption of denial**" for license applications, effectively eliminating access to U.S. technology.

- The policy targets firms involved in:
  - National security risks
  - Military-civil fusion
  - Diversion of dual-use technologies

Research Question: Do US export controls have a measurable impact on target firm performance?

Objective

Estimate *causal impacts* of U.S. economic policies (**BIS Entity List**) on PRC firms.

Approach

- Merge commercially available data that covers both public and private firms with listings on the US federal register to construct the largest possible sample.
- When enough data exists, utilize common DiD estimators for staggered treatments (Callaway & Sant'Anna, 2021)
- If insufficient data for estimators that rely on large samples, use permutation based methods (i.e. results by industry).

Related Literature

Economic Sanctions on Firm Performance

Ahn and Ludema (2020)
Gaur et al. (2023)
Keerati (2022)
Nigmatulina (2025)
Gullstrand (2020)
Görg et al. (2024)
Egorov et al. (2025)
Basker and Kamal (2025)

Export Controls in US-China Competition

Crosignani et al. (2025)
Liu et al. (2024)
Shen et al. (2024)
Anwar et al. (2024)
He et al. (2025)
Hu et al. (2024)
He and Lyu (2025)
Alfaro et al. (2025)
Huang et al. (2025)
Liu et al. (2025)

Economic Statecraft & Geoeconomics

Drezner (2003, 2024)
Morgan et al. (2009)
Morgan et al. (2023)
Felbermayr et al. (2020)
Clayton et al. (2025)

This paper contributes first comprehensive firm-level analysis of BIS Entity List impacts on financial performance of PRC firms

Data: Moody's Orbis Database & Sample Construction

Orbis Database (Bureau van Dijk/Moody's)

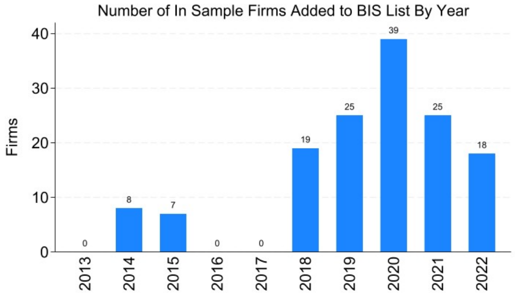
- Global firm-level database: financial statements and ownership
- **China coverage:** $\sim$3 million firms with some data
- Sources: Credit rating agencies, tax filings, national databases, public records

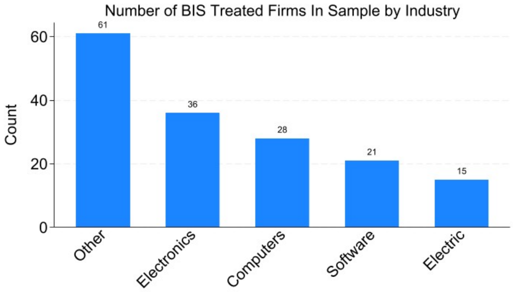
Sample Construction

- **BIS Entity List:** 1,227 Chinese firms from Consolidated Screening List
- **Matching process:** Multi-step linking using English/Chinese names, Tax ID, Unified Social Credit Code (USCC); 660 matched firms, 194 firms with some financial data

Data: Sample Construction - Cont.

- **Final treated sample:** 149 firms (sufficient data: $\geq$5 years, $\geq$3 post-treatment observations.)
- **Control sample:** 115,333 untreated firms from 57 industries
- Unbalanced panel spanning **2012–2024** with annual observations





Empirical Strategy

Estimating the Average Effect of the Entity List on PRC Firms

- Problem: Timing of treatment varies by firm
- Solution: *Callaway and Sant'Anna (2021)* Difference in Differences (CSDiD)

Estimating Heterogeneous Impacts of the Entity List Across Sectors

- Problem: Too few treated firms for Sector subgroups analyses using CSDiD
- Solution: *Permutation-based estimator* (Heckman et al., 2010) with K-Nearest Neighbors (KNN)

Callaway & Sant'Anna (2021) Difference-in-Differences

Why CSDiD? – *Staggered treatment:* Firms added to Entity List in different years

- Only include listed firms; *never-treated group* are more recently listed firms
- **Cohort-time effects:** $ATT(g, t)$ for firms first treated in period g, evaluated at time t
- **Valid comparisons only:** Uses firms untreated at time t as controls
- **Aggregation:** Event-time dynamics via weighted average across cohorts

$$\theta(k) = \sum_g w_{g,k} \cdot ATT(g, g+k), \quad \sum_g w_{g,k} = 1$$

where $k \equiv t - g$ is the years since treatment.

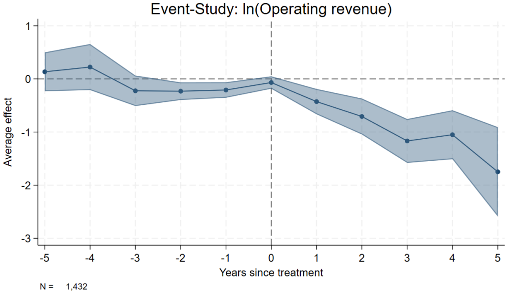
Main Findings: Average Treatment Effects (CSDiD)

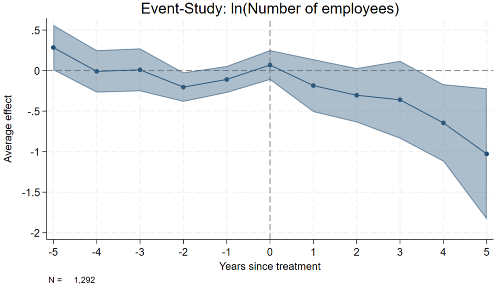
Outcome Variable	Magnitude	ATE	S.E.
Strong Negative Effects:			
Operating Revenue	-58%	-0.86^{***}	(0.148)
Intangible Assets	-61%	-0.95^{***}	(0.317)
Cash Assets	-51%	-0.72^{***}	(0.236)
Employment	-33%	-0.41^{**}	(0.166)
Total Assets	-36%	-0.45^{***}	(0.177)
Cost of Goods Sold	-44%	-0.58^{***}	(0.166)
Current Liabilities	-42%	-0.58^{***}	(0.145)
No Significant Effect:			
Solvency Ratio	-7%	-0.074	(0.146)
Current Ratio	2%	0.02	(0.126)

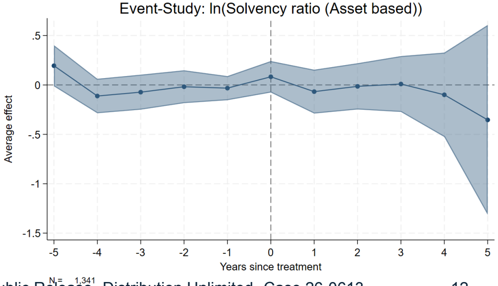
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. 5-year post-treatment horizon.

Entity List causes firms to shrink substantially but does not significantly impact solvency measures

CSDiD ATE

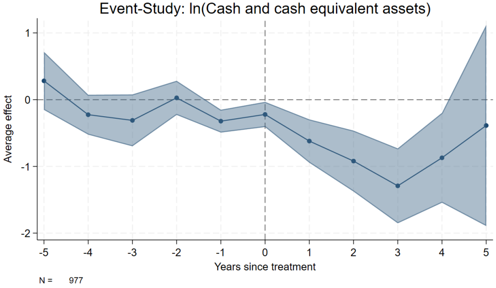


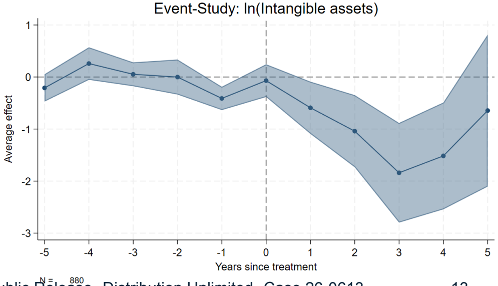




- Revenue ⇓ – Possible chilling effect on firm demand
- Employment ⇓ – Falling over time but at a slower pace than the fall in revenue
- Solvency Ratio shows no response post treatment

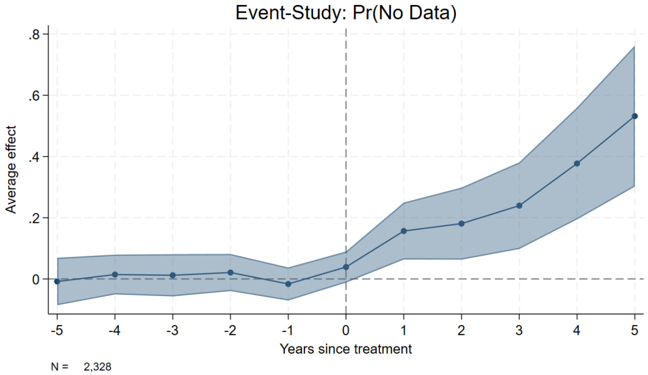
CSDiD ATE (Cont.)





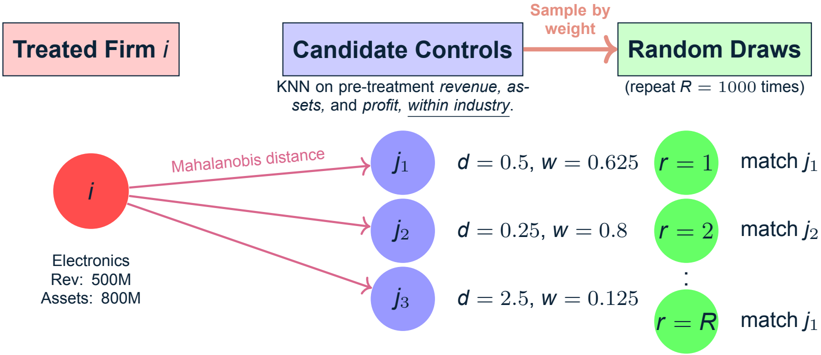
- Intangible Asset Value ⇓ – loss of US tech. licensing and key suppliers
- Cash Assets ⇓ – using liquid assets to cover liabilities
- Total Assets ⇓ – No measurable change in fixed assets

Do Firms Survive?



- **Almost no firms exit the market during our sample**, but we do observe a large reduction in firms that report data to Moody's post treatment.
- **This could capture firms folding** or it could be firms turning to domestic channels for debt financing.

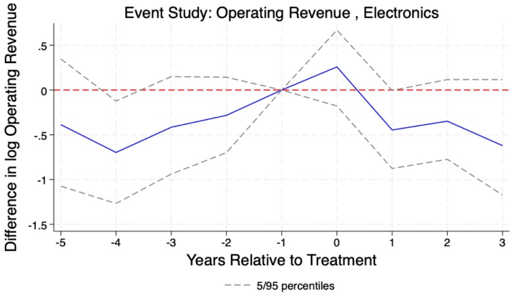
Permutation-Based Matching: Visual Intuition

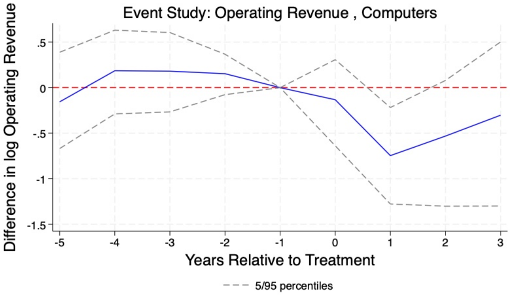


Weights: $w_{ij} = \frac{1/d_{ij}}{\sum_{j \in KNN(i)} (1/d_{ij})}$

Average: $\hat{\beta}_h^{MC} = \frac{1}{R} \sum_{r=1}^{R} \hat{\beta}_h^{(r)}$
with 5th/95th percentiles

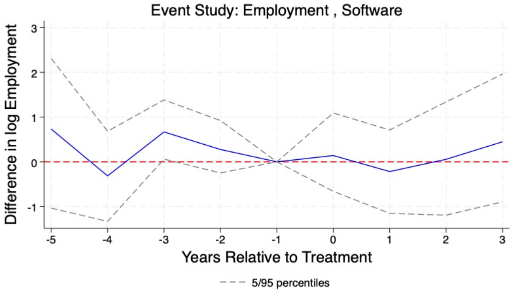
Permutation-Based Results: Operating Revenue

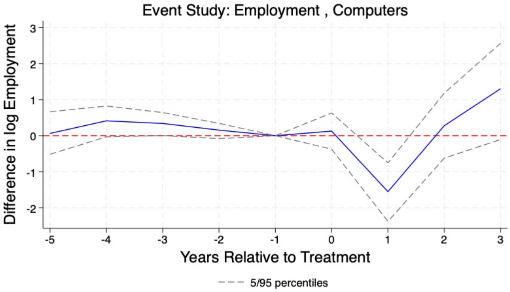




- **Revenue declines across all industries**, but Electronics and Computer sectors experience the fastest and most pronounced drops.

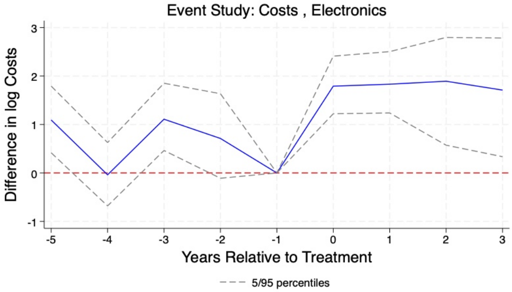
Permutation-Based Results: Employment

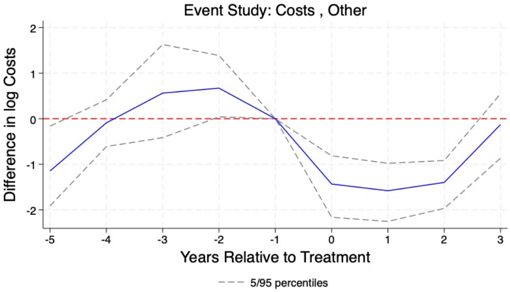




- **Employment effects are heterogeneous**: Computer industry firms show immediate employment losses tracking revenue declines.

Permutation-Based Results: Costs





- **Cost (COGS) responses differ by sector**: Electronics firms see costs *increase* after treatment.

Is trade/business rerouted through unlisted firms?

- This would bias our results as our control group would see a positive effect on financials after a firm is treated.
- To test this, we run a two way fixed effects regression for the share of the industry that receives treatment on financial outcomes.

$$log(x_{it}) = \beta Exposure_{s(i),t} + \iota_i + \rho_t + \varepsilon_{it}$$

where $Exposure_{s(i),t}$ is the share of firms in industry s (that firm i belongs to) that are on the BIS Entity List in year t.

Outcome (log)	Exposure	N
Operating Revenue	-0.0321 (0.0454)	3,139,097
Current Ratio	0.0321 (0.0216)	1,133,277
Solvency Ratio	0.0175 (0.0140)	2,211,072
Employees	-0.0049 (0.0266)	1,788,804
Intangible Assets	0.0516 (0.0454)	267,620
Cash & Equivalents	0.0712 (0.0620)	1,166,897
Current Liabilities	0.0364* (0.0214)	1,165,812
Cost of Goods Sold	0.2212** (0.0911)	1,673,414
Operating Profit (EBIT)	0.0543 (0.0741)	1,186,447

Notes: SEs in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Conclusions

1. Export controls impose real financial costs on targeted firms.
   - BIS Entity List designations significantly reduce revenue, employment, and assets of targeted Chinese firms.
2. Export controls remain effective over the long run.
   - Most effects persist and grow over time, suggesting evasion is costly or difficult.
3. Export controls are most effective in industries where the US has significant technological advantage.
   - Computer and electronic sectors experience higher costs and larger declines in sales compared to other sectors.
4. Firms are much less likely to participate in international debt markets after treatment.
   - Currently unable to identify if firms are exiting or relying on domestic markets/government support for financing needs.

Questions?

Thank you

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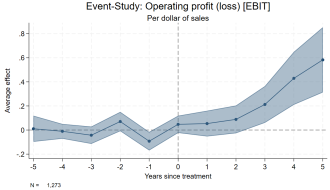
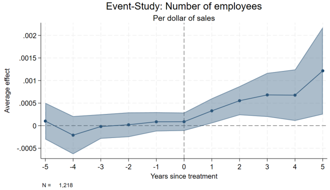
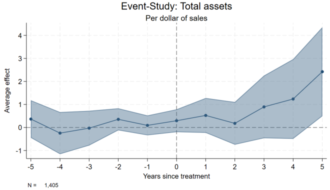
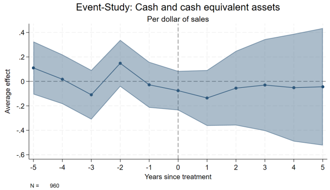
Backups

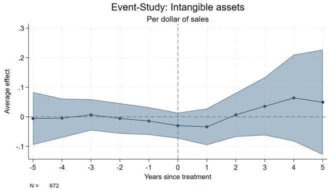
Results as a Share of Revenue

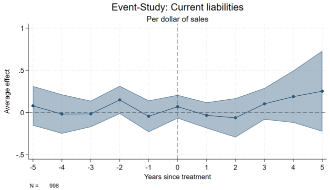












Robustness: Permutation-Based ATE Results